

Linux Basics Command

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1. `ls`
 - list the files and directories in the current directories. Ex: `ls [option]`
[directory_or_path] option :
(-a) to show the hidden content.
(`ls /home` to see the created file or directory under home directory)

```
root@6746e12274bb88419b4b5f99:/# ls
bin boot code-server_4.18.0_amd64.deb data dev etc home lib lib32 lib64 libx32 mariadb_repo_setup media mnt proc root run sbin snap srv sys usr var
```

2. `cd`
 - change the current directories. Ex. `cd dir_name`
`cd` : returns to the current user's home directory.
`cd..` : moves a directory up
`Cd-` : goes back to the previous directory

```
root@6746e12274bb88419b4b5f99:/# cd mnt
root@6746e12274bb88419b4b5f99:/mnt# cd ..
root@6746e12274bb88419b4b5f99:/#
```

3. `pwd`
 - to check the full path of your current/present working directory. Ex. `pwd [option]`
-L : the -L option prints environment variable
-P : meanwhile, -p outputs the exact location.

```
root@6746e12274bb88419b4b5f99:/mnt# pwd -LP
/mnt
root@6746e12274bb88419b4b5f99:/mnt# pwd -P
/mnt
root@6746e12274bb88419b4b5f99:/mnt# pwd -L
/mnt
```

4. `sudo -i`
 - Superuser do or sudo enables non-root users who are part .of the sudo group to execute administrative commands. Ex. `sudo -i`

```
labex:project/ $ sudo -i
root@6748288274bb88419b4bd2b3:~#
```

5. `su`
 - The su command lets you switch to another user in the terminal session. Ex. `su user_name`

```
[ec2-user@ip-172-31-3-81 ~]$ sudo su
[root@ip-172-31-3-81 ec2-user]#
```

6. `whoami`
 - You can check the currently logged-in user from the Linux command-line shell Ex. `whoami`

```
root@6748288274bb88419b4bd2b3:~# whoami
root
```

7. `clear`
 - To use the clear the screen Ex. `clear` shortcut: `ctrl + L`

8. `history`
 - To show the previous command history in control panel Ex. `History`

```
root@6748288274bb88419b4bd2b3:~# history
1 su home
2 su local
3 whoami
4 clear
5 history
```

9. `hostname`
 - To show name of the system Ex. `Hostname`

```
root@6748288274bb88419b4bd2b3:~# hostname
6748288274bb88419b4bd2b3
```

```
[root@ip-172-31-3-81 ec2-user]# hostname
ip-172-31-3-81.ap-south-1.compute.internal
[root@ip-172-31-3-81 ec2-user]# hostname linux
linux
[root@ip-172-31-3-81 ec2-user]#
```

10. `cal`
 - To show the calendar Ex. `cal -3` (show the 3 months) `cal -2025` (show the hole 2025 calendar)

```
[root@ip-172-31-3-81 ~]# cal -3 2025
      October 2025      November 2025      December 2025
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1  2  3  4  5  6  7  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
```

```
[root@ip-172-31-3-81 ~]# cal -3 2025
      October 2025      November 2025      December 2025
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4           1               1  2  3  4  5  6
 5  6  7  8  9 10 11   2  3  4  5  6  7  8   7  8  9 10 11 12 13
12 13 14 15 16 17 18   9 10 11 12 13 14 15  14 15 16 17 18 19 20
19 20 21 22 23 24 25  16 17 18 19 20 21 22  21 22 23 24 25 26 27
26 27 28 29 30 31     23 24 25 26 27 28 29  28 29 30 31
                        30
```

11. uname

- To show the user or system information Ex. uname

```
root@6748288274bb88419b4bd2b3:~# uname
Linux
```

12. date

- To show the date Ex. date

```
root@6748288274bb88419b4bd2b3:~# date
Thu Nov 28 16:31:10 CST 2024
```

13. tty

- This command displays current user terminal number.

```
[root@ip-172-31-3-81 ec2-user]# tty
/dev/pts/2
[root@ip-172-31-3-81 ec2-user]#
```

14. w

- Display all current user details with their time schedule.

```
[root@ip-172-31-3-81 ~]# w
05:54:32 up 32 min, 3 users, load average: 0.00, 0.00, 0.00
USER      TTY      LOGIN@  IDLE   JCPU   PCPU WHAT
ec2-user pts/0    05:36   17:38   0.01s  0.01s -bash
ec2-user pts/1    05:52   1:28   0.01s  0.02s sshd: ec2-user [priv]
ec2-user pts/2    05:53   0.00s  0.01s  0.01s sudo -i
```

15. uname -a

- It displays operating system kernel name.

```
[root@ip-172-31-3-81 ~]# uname -a
Linux ip-172-31-3-81.ap-south-1.compute.internal 6.1.115-126.197.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Nov 5 17:36:57 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
```

16. who

- Display current logged in all users with their details.

```
[root@ip-172-31-3-81 ~]# who
ec2-user pts/0    2024-11-30 05:36 (152.58.33.172)
ec2-user pts/1    2024-11-30 05:52 (152.58.33.172)
ec2-user pts/2    2024-11-30 05:53 (152.58.33.172)
```

17. uname -r

```
[ec2-user@ip-172-31-3-81 ~]$ uname -r
6.1.115-126.197.amzn2023.x86_64
```

18. free -h

- It display RAM information in human readable form

```
[root@ip-172-31-3-81 ec2-user]# free -h
              total        used        free      shared  buff/cache   available
Mem:           949Mi       129Mi       589Mi         0Ki       230Mi       682Mi
Swap:            0B           0B           0B
[root@ip-172-31-3-81 ec2-user]#
```

19. which

- To know path of command file.

```
[root@ip-172-31-3-81 ec2-user]# which mkdir
/usr/bin/mkdir
[root@ip-172-31-3-81 ec2-user]#
```

20. wc

- Word count

```
[root@ip-172-31-3-81 ec2-user]# cat > data.txt
hii
bay
welcome
aaaaa
exit
^Z
[1]+  Stopped                  cat > data.txt
[root@ip-172-31-3-81 ec2-user]# wc data.txt
 5  5 27 data.txt
```

21. alias

- Command (alias 't'='touch')
- alias (show the all alias)
- unalias t (remove the alias)

```

root@ip-172-31-6-179:~# alias 't='touch'
root@ip-172-31-6-179:~# touch file.txt
root@ip-172-31-6-179:~# ls
file.txt  snap
root@ip-172-31-6-179:~# t data.txt
root@ip-172-31-6-179:~# ls
data.txt  file.txt  snap
root@ip-172-31-6-179:~# alias
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
alias t='touch'
root@ip-172-31-6-179:~# unalias t
root@ip-172-31-6-179:~# alias
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'

```

22. Alias permanent

23. tree

- To show the data in tree format

```

root@ip-172-31-4-5:~# mkdir -p /root/project/demo && touch /root/project/file.txt
root@ip-172-31-4-5:~# tree /root
/root
├── file.txt
├── project
│   └── demo
│       └── file.txt
├── snap
│   └── amazon-ssm-agent
│       ├── 9881
│       ├── common
│       └── current -> 9881
└── 8 directories, 2 files

```

24. hostnamectl

- Show detailed information about system

25. locate

- They are using in company
- Locate faster than find command
- The locate command relies on a database file (mlocate.db) that stores a list of file paths.
- When you run locate, it quickly searches the database instead of scanning the actual filesystem.
- **Common Options**

Option	Description
-i	Perform a case-insensitive search.
-r [pattern]	Use a regular expression for the search.
-l [number]	Limit the output to the specified number of results.
-c	Count the number of matching entries instead of listing them.
-- existing	Only list files that currently exist on the filesystem (ignores deleted files).

26. find

- Is slower than the locate command
- They are searching whole system
- But they give the more option
- They are no need to updatedb

27. file

- This command are to used show the file type

28. sad

29. awk

30. tee

31. which

Correct Usage of the source Command

The source command is used to load or re-execute a script file, such as .bashrc, into the current shell session. It is generally used to apply changes made to a configuration file immediately.

Key Errors in Your Commands

1. Misspelling of source:

The command you used, souchr, is incorrect. It should be source.

2. Incorrect File Paths:

- o You added unnecessary spaces, e.g., /root/.bashrc instead of /root/.bashrc.
- o Passing /root/ as an argument to source is invalid because it's a directory.

3. Invalid Commands:

- o `t file23.txt` is not a valid command. You likely intended to create or view `file23.txt`.

Fixes for Each Issue

1. Correctly Source `.bashrc`

- To reload the `.bashrc` file in the root user's home directory:
`source /root/.bashrc`
- Alternatively, while logged in as root:
`source ~/.bashrc`
- If you're a regular user and need to source your `.bashrc`:
`source ~/.bashrc`

2. Resolve `t file23.txt`

- If you want to create `file23.txt`, use:
`touch file23.txt`
- If you want to view its contents:
`cat file23.txt`

3. Avoid Adding Spaces in File Paths

- Example of incorrect usage:
`source /root /.bashrc` # Invalid: Extra space.
- Correct it to:
`source /root/.bashrc`

General Notes

1. If `.bashrc` Does Not Exist:

Verify that the `.bashrc` file exists in the intended location:

```
ls -la ~/.bashrc
```

If it doesn't exist, you might need to create or copy it from a default configuration:

```
cp /etc/skel/.bashrc ~/.bashrc
source ~/.bashrc
```

2. Loading Configuration for Root and Non-Root Users:

Ensure you're sourcing the correct `.bashrc` file based on the user.

Example Workflow

```
# Navigate to root's home directory.
```

```
cd /root
```

```
# Verify if .bashrc exists.
```

```
ls -la .bashrc
```

```
# Source .bashrc correctly.
```

```
source /root/.bashrc
```